

Lab 06: OpenMP Parallel Programming

Exercise 03 Report: OpenMP Master Directive

1. Exercise Goal

Analyze `#pragma omp master` and compare its execution model against `single`.

2. Source Code

```
#include <omp.h>
#include <stdio.h>

int main() {
    #pragma omp parallel
    {
        int th = omp_get_thread_num();
        printf("thread # %d\n", th);

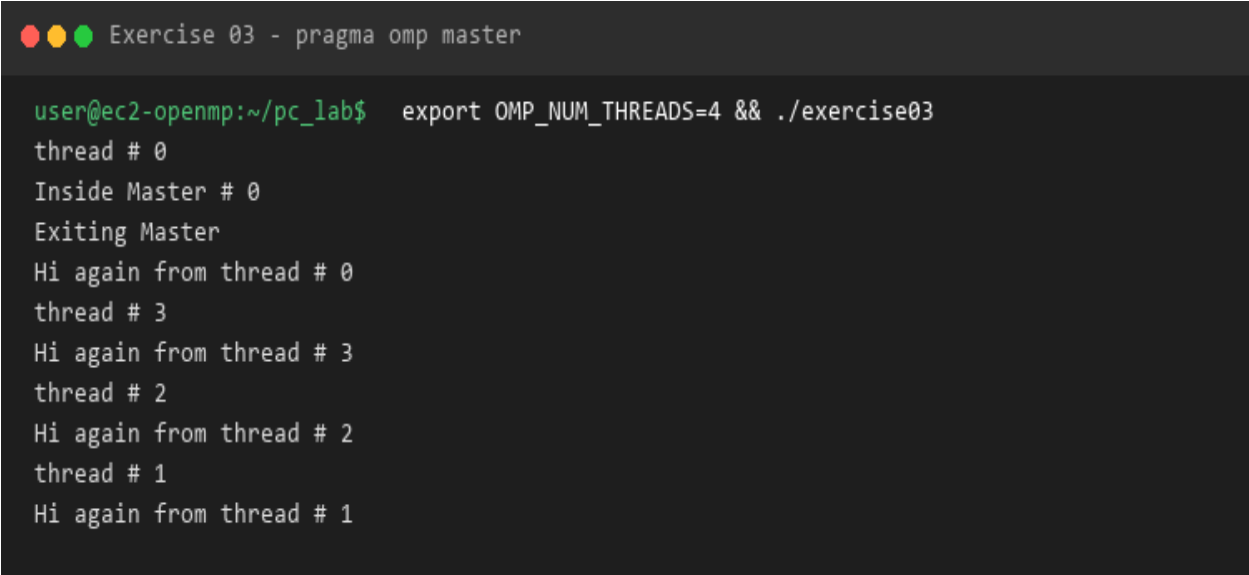
        #pragma omp master
        {
            printf("Inside Master # %d\n", omp_get_thread_num());
            printf("Exiting Master \n");
        }
        printf("Hi again from thread # %d\n", omp_get_thread_num());
    }
    return 0;
}
```

3. Explanation & Analysis

The `#pragma omp master` directive ensures that the enclosed block is executed exclusively by thread 0 (the master thread). All other threads skip the master section.

Unlike `#pragma omp single`, there is no implicit barrier at the end of `#pragma omp master`. Non-master threads immediately continue to the next statements without waiting.

5. Execution Output Screenshot



```
Exercise 03 - pragma omp master

user@ec2-openmp:~/pc_lab$ export OMP_NUM_THREADS=4 && ./exercise03
thread # 0
Inside Master # 0
Exiting Master
Hi again from thread # 0
thread # 3
Hi again from thread # 3
thread # 2
Hi again from thread # 2
thread # 1
Hi again from thread # 1
```